



Technical Campus

योग: कर्मसु कौशलम्
IN PURSUIT OF PERFECTION



**SCHOOL OF ENGINEERING
AND TECHNOLOGY**

Process Automation (IOT310)

By: Dr. Divya Agarwal

Course Overview:

- Process and Industrial Automation specialization offers comprehensive knowledge and professional level skills focused on developing and deploying software robots. It starts with the basic concepts of Robotic Process Automation. In the present global scenario, industries are moving towards complete automation. Hence, this course is foundation of engineers who want to specialize in industrial and process automation field.

Course Objective:

- To understand the Fundamental and basic principles of Robotics Process Automation, Applications in various Industries.
- To install the UiPath Studio and understand difference between different versions
- To use different activities while using excel sheet and automate email related activities.
- To understand benefits of given industrial automation systems, describe their functions and compare characteristics of given automation systems.

Course Outcomes (CO):

- **CO1** Able to Map and assess some of the business processes that are fit for automation.
- **CO2** Able to connect to UiPath Automation Cloud, download the installer, and set up their own attended automation environment and Build an automation using StudioX.
- **CO3** Able to effectively automate tasks involving use of Excel files, adapt the automation project to dynamic Excel ranges.
- **CO4** Able to apply and map realtime application of industrial automation

Syllabus Contents



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- **UNIT IV**
- **Introduction to Industrial Automation:** Need and benefits of Industrial Automation, Automation Hierarchy, Basic components of Automation systems, description of each component, types of automation systems flexible, fixed and programmable systems
- **The Future Automated Factory:** Future Automated Factory, Human Workers in the Future Automated Factory, The social impact

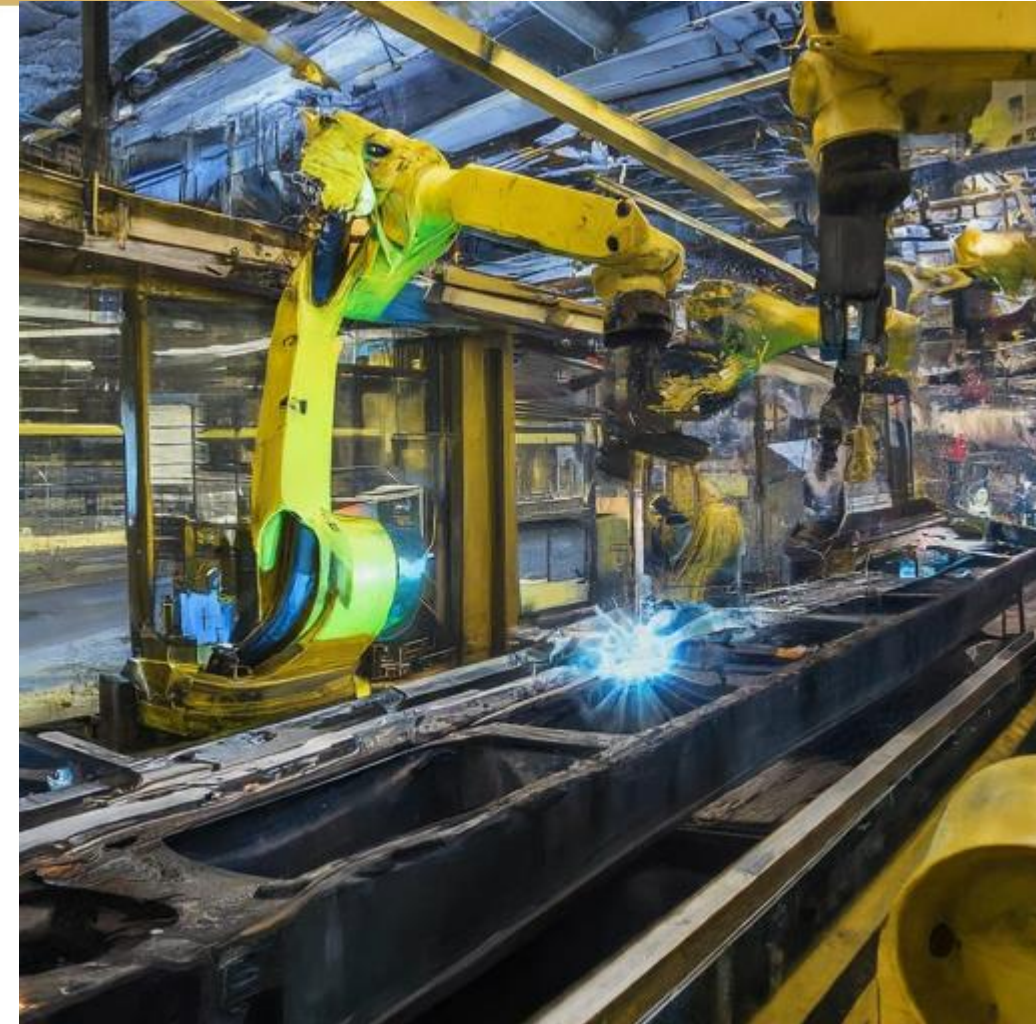
Introduction to Industrial Automation

- Use of technology to operate machines and processes with minimal human intervention.
- Employs sensors, actuators, and programmable controllers.
- Automates repetitive tasks, decision-making, and physical operations.
- Refers to transformation of manual industrial processes into automated ones significantly reducing human involvement.



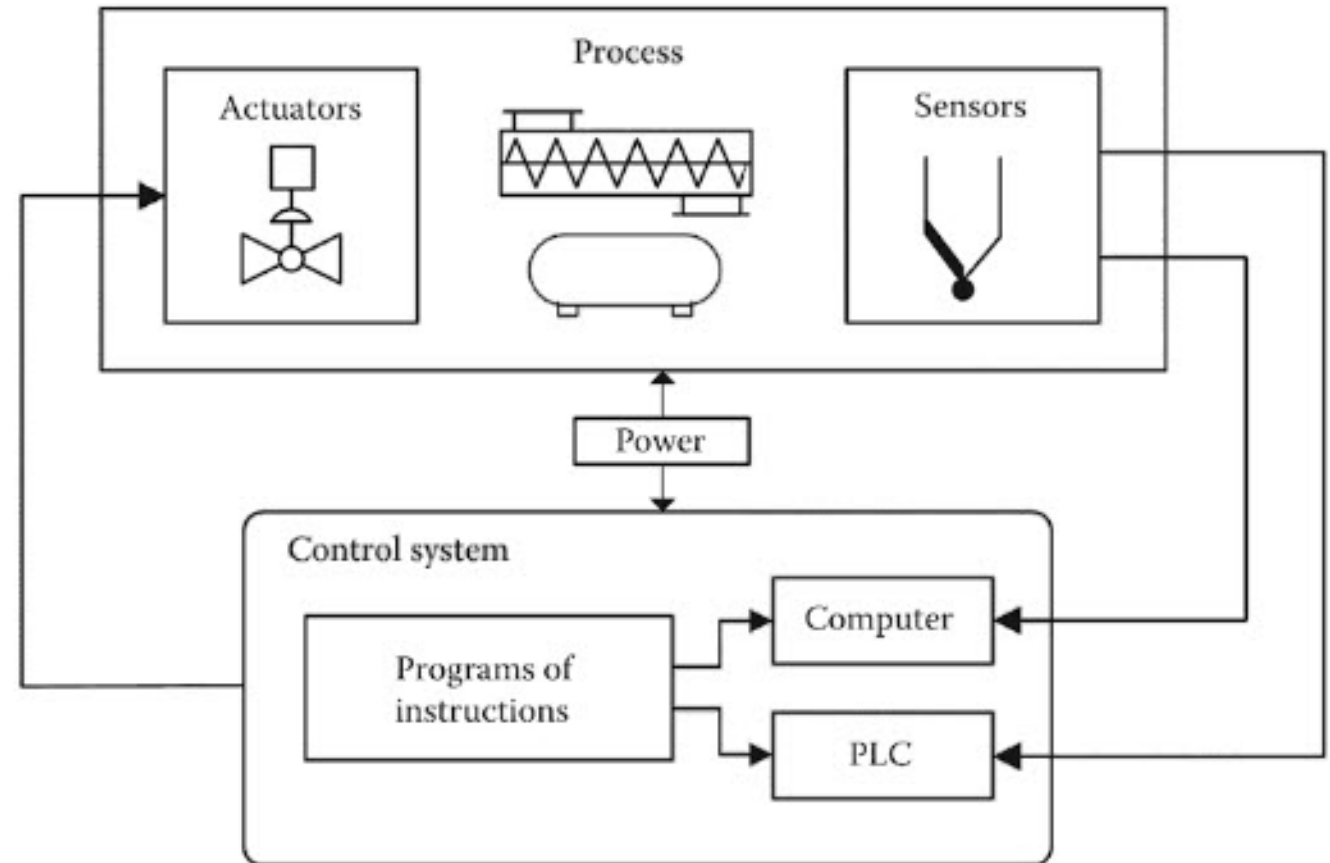
Introduction to Industrial Automation

- Industrial Automation is the use of control systems, including machines, actuators, sensors, processors, and networks to perform tasks, with the goal of automating production.
- Its history started with simple conveyor belts pulling parts through an assembly line.
- Machines performed basic tasks and work that reduced manual labor.
- It encompasses wide range of machines, actuators, sensors, processors, and networks that work to [connect an industrial environment](#).
- From PLCs, AI, Machine Learning, and IIoT devices, modern industrial automation is focusing on the best way to leverage technology.
- Following will introduce industrial automation, its types, and its benefits that are leading global industries, expanding markets, and evolving competition.



Significance of Industrial Automation

- Industrial Automation plays a pivotal role in modern industries
- Boosts production output and overall efficiency by streamlining processes and reducing downtime.
- Enhances product quality and consistency by eliminating human error.
- Improves workplace safety by reducing human exposure to risks.
- Lowers manufacturing costs and creates a competitive edge.



Historical Perspective: Evolution of Automation

- Early Automation (18th Century):
 - **Industrial Revolution** - Rise of steam and water-powered machines, significantly increasing production capacity.
 - **Interchangeable parts** - Pioneered by Eli Whitney in firearms production which allowed for faster and more consistent assembly.
- Mid-20th Century:
 - **Assembly lines** - Mass production techniques pioneered by Henry Ford.
 - Revolutionized car manufacturing by breaking down complex tasks into simpler, repetitive steps performed by workers along a moving conveyor belt.
 - **Transfer machines** - Automated workstations for specific tasks.

Historical Perspective: Evolution of Automation

- Late 20th Century: Witnessed a paradigm shift with the introduction of Programmable Logic Controllers (PLCs)
 - **Programmable Logic Controllers (PLCs)** - Introduced for industrial control applications enabling flexible and sophisticated automation programs..
 - **Industrial robots** - Increased use for complex tasks like welding, painting, and material handling.

Need for Industrial Automation in Modern Manufacturing

- **Global competition** - Demands increased production efficiency and cost reduction to stay ahead.
- **Labor costs** - Rising labor costs necessitate automation for economic viability.
- **Product complexity** - Modern products are often intricate and require precise and consistent manufacturing processes.
- **Safety concerns** - Automation can remove humans from hazardous environments.

Benefits of Industrial Automation

- Industrial Automation is rapidly transforming the landscape of modern manufacturing.
- It offers a compelling array of benefits, significantly boosting productivity, efficiency, safety, and quality control.
- As automation technologies continue to evolve and become more sophisticated, their role in industrial processes will only become more prominent.
- By embracing automation, manufacturers can gain a significant competitive edge in the global marketplace.

Increased Productivity -

- Increased production output through faster cycle times and 24/7 operation maximizing production time, compared to manual processes.
- Automation also reduces downtime by minimizing equipment breakdowns through preventive maintenance and faster troubleshooting.
- Eliminates human fatigue and inconsistencies, leading to a steadier and more predictable production flow.

Increased Efficiency –

- Industrial Automation is a champion of efficiency.
- Optimized use of resources like raw materials, energy, and labor, minimizing waste.
- Streamlined production processes, reducing the need for rework due to human error.
- Automation enables collection of real-time data from various stages of production process. This data can be analyzed to identify areas for improvement and continuously optimize entire manufacturing system.

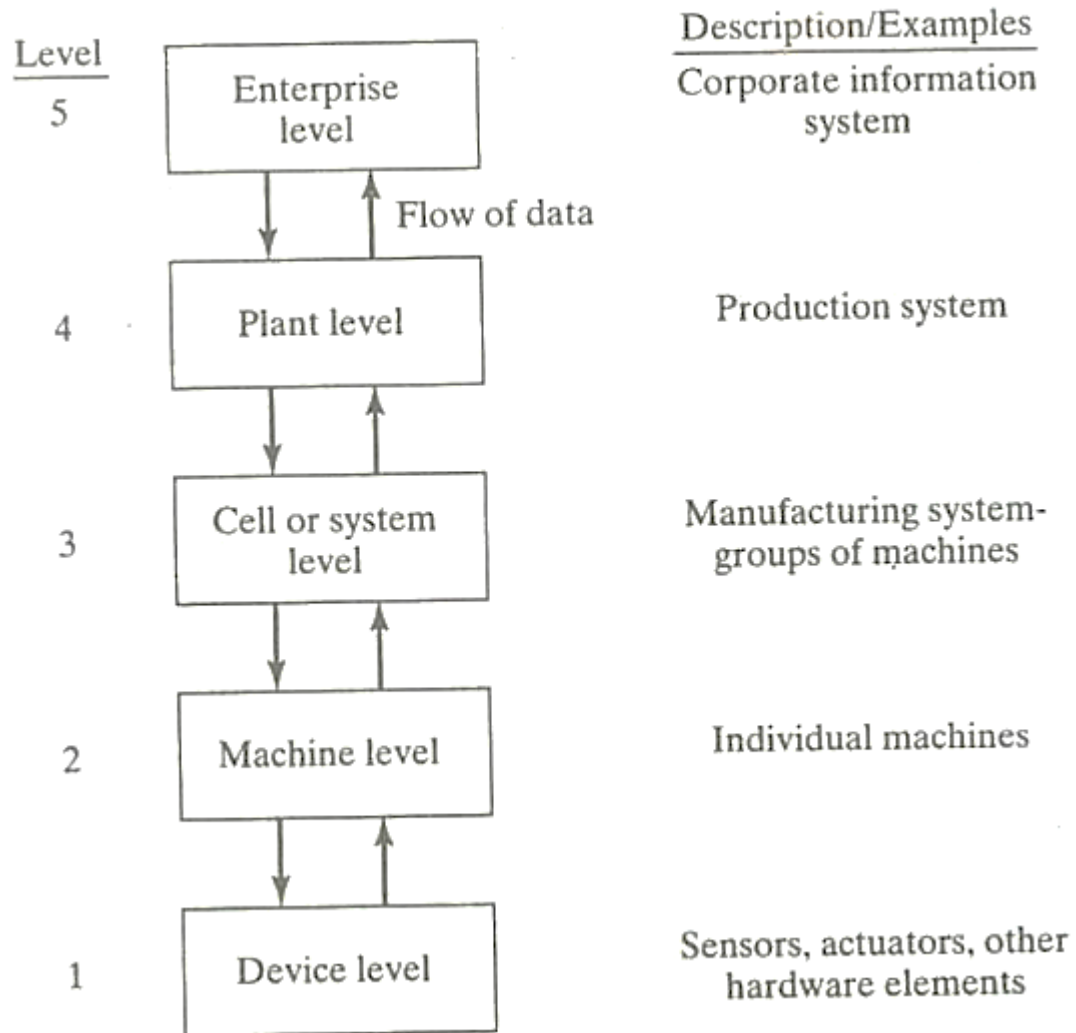
Improved Safety

- Safety is a paramount concern in any industrial setting.
- Industrial Automation plays a vital role in enhancing workplace safety.
- By automating hazardous tasks like welding, painting, and working with heavy machinery, risk of accidents involving human workers is significantly reduced.
- Automation leads to a cleaner and safer work environment through improved air quality control and reduced noise levels.
- Automated systems can be integrated with safety protocols like emergency shut-off mechanisms and automated alerts, further safeguarding workers and equipment.

Enhanced Quality Control

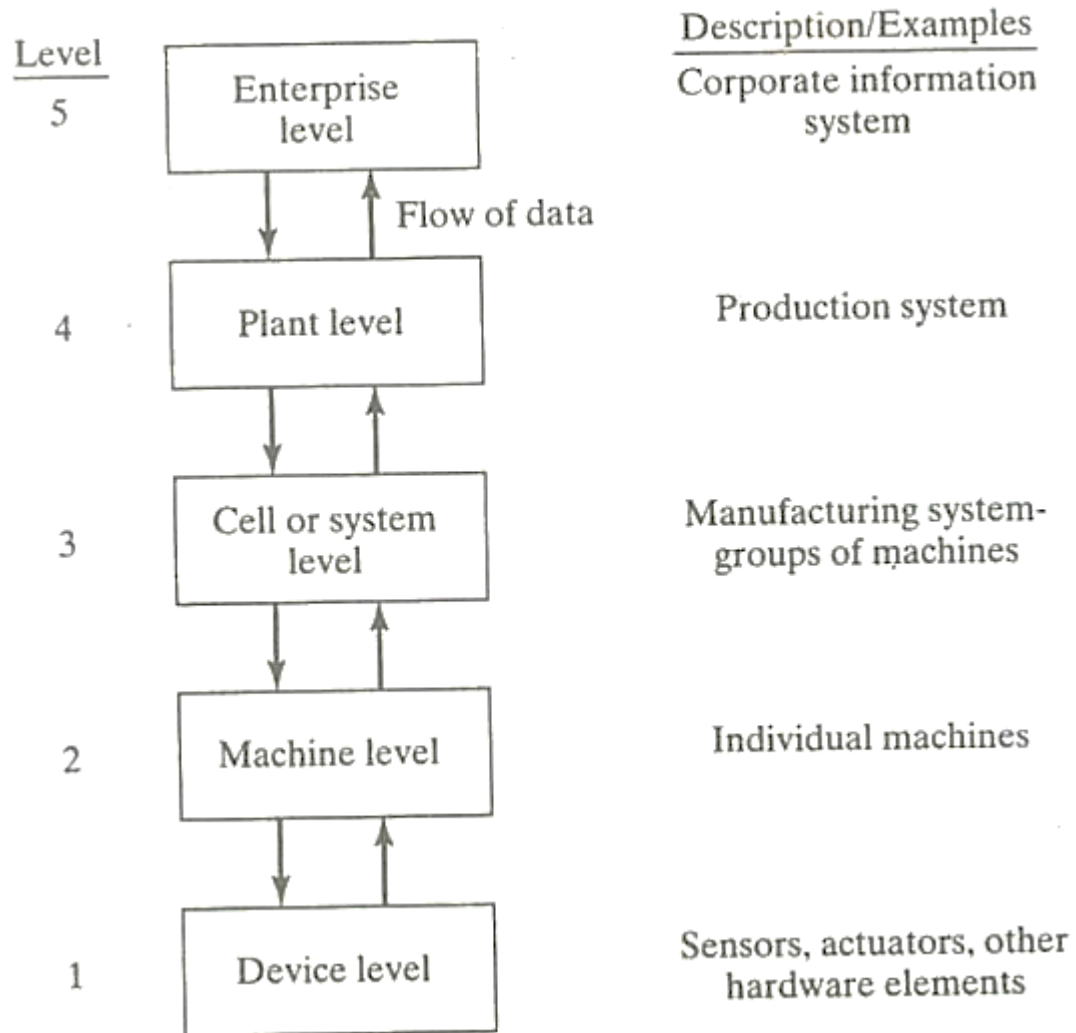
- Automated systems perform tasks with exceptional precision and repeatability, leading to consistently high-quality products.
- Automated inspection systems can be integrated into production line to ensure every product meets strict quality standards.
- Real-time data collected from sensors and inspection systems allows for early detection and correction of defects, minimizing the production of faulty items.

Hierarchical Structure of Industrial Automation



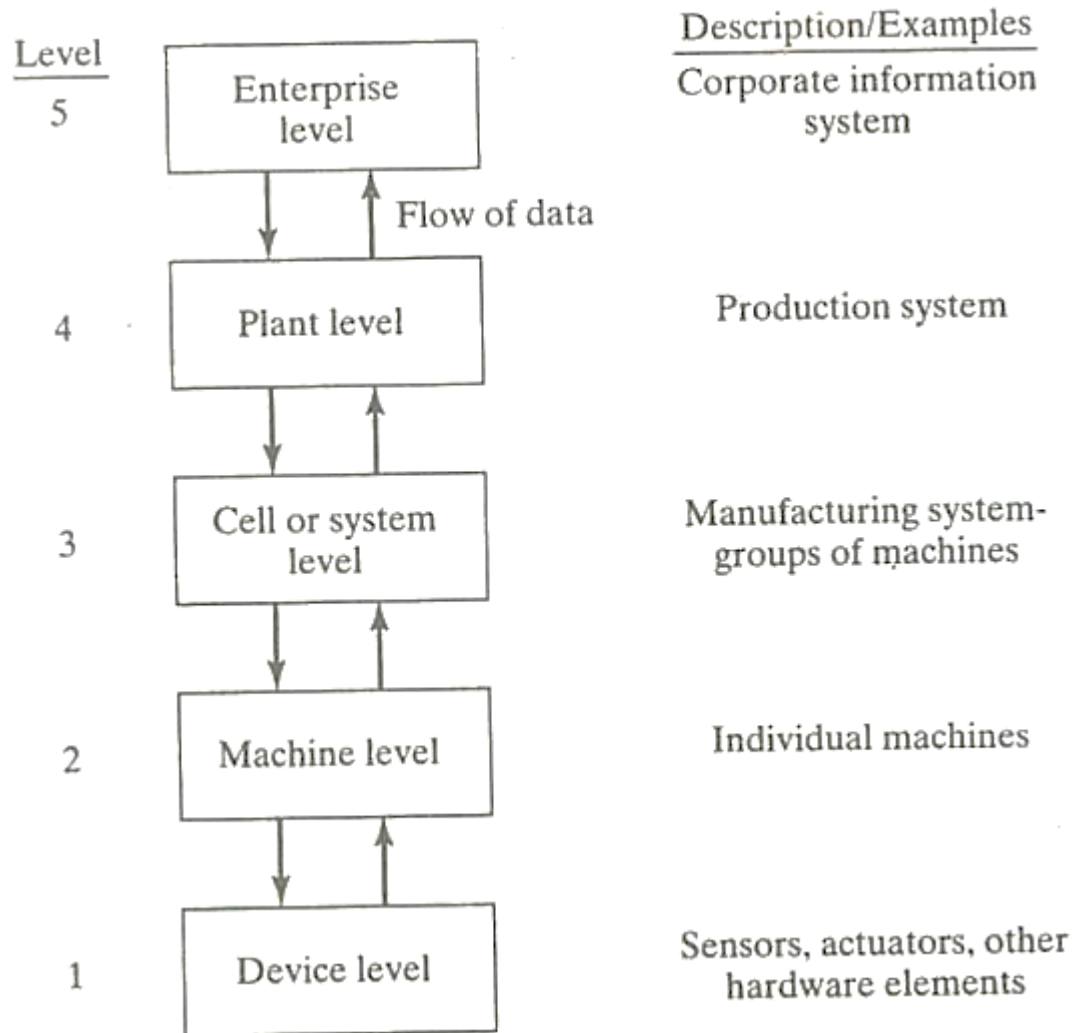
- **Level 1:** Introduces basic automation elements like sensors and actuators.
- **Functionality:**
 - Introduces basic automation elements like sensors and actuators.
 - Sensors monitor process variables like temperature, pressure, or flow.
 - Actuators respond to sensor readings by making simple adjustments.
 - Limited decision-making capabilities.
- **Example:** A pressure relief valve that automatically opens to release pressure if a sensor detects a dangerous increase.

Hierarchical Structure of Industrial Automation



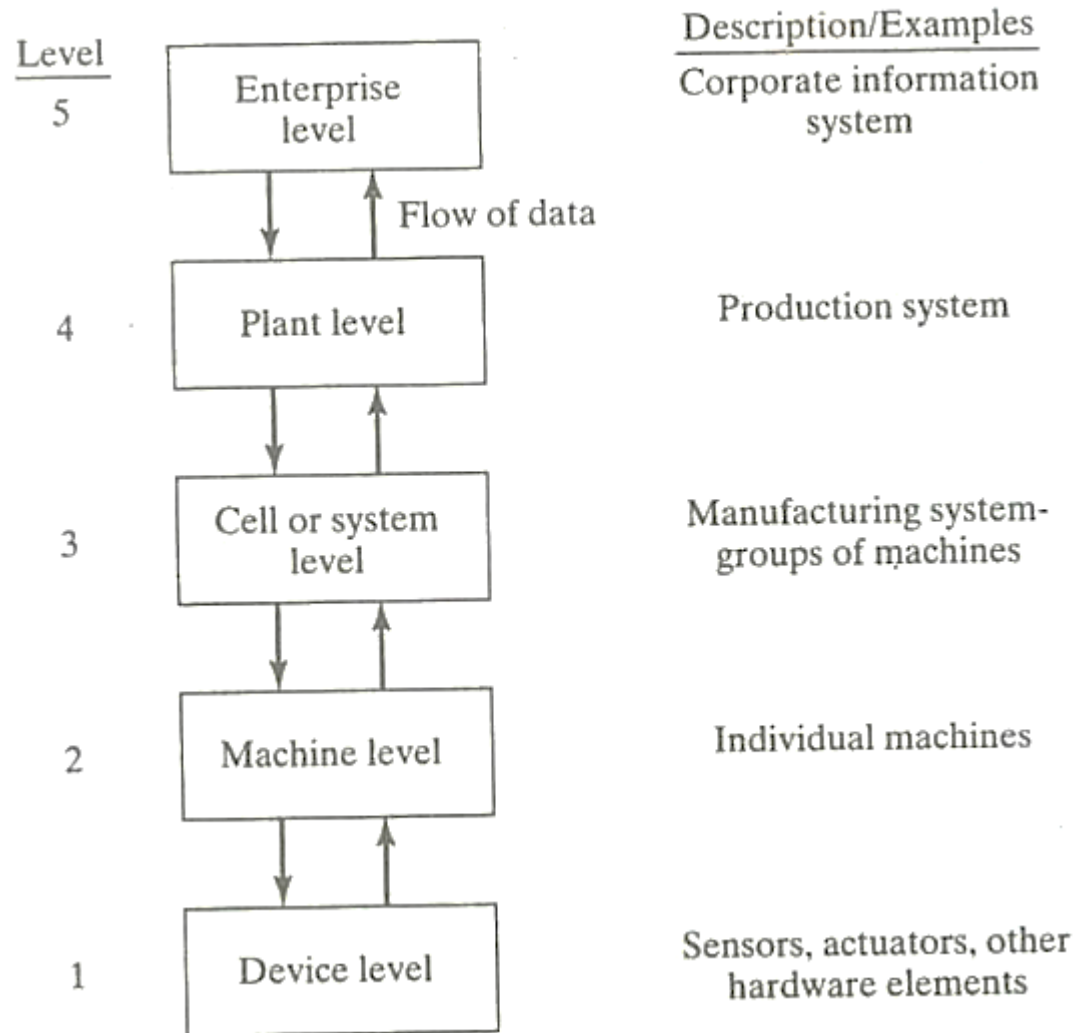
- **Level 2: Machine level.**
- Hardware at device level is assembled into individual machines.
- Examples include CNC machine tools and similar production equipment, industrial robots, powered conveyors, and automated guided vehicles.
- Control functions at this level include performing the sequence of steps in the program of instructions in the correct order and making sure that each step is properly executed.
- **Example:** A PLC controlling a conveyor belt system, adjusting speed and direction based on product arrival and destination.

Hierarchical Structure of Industrial Automation



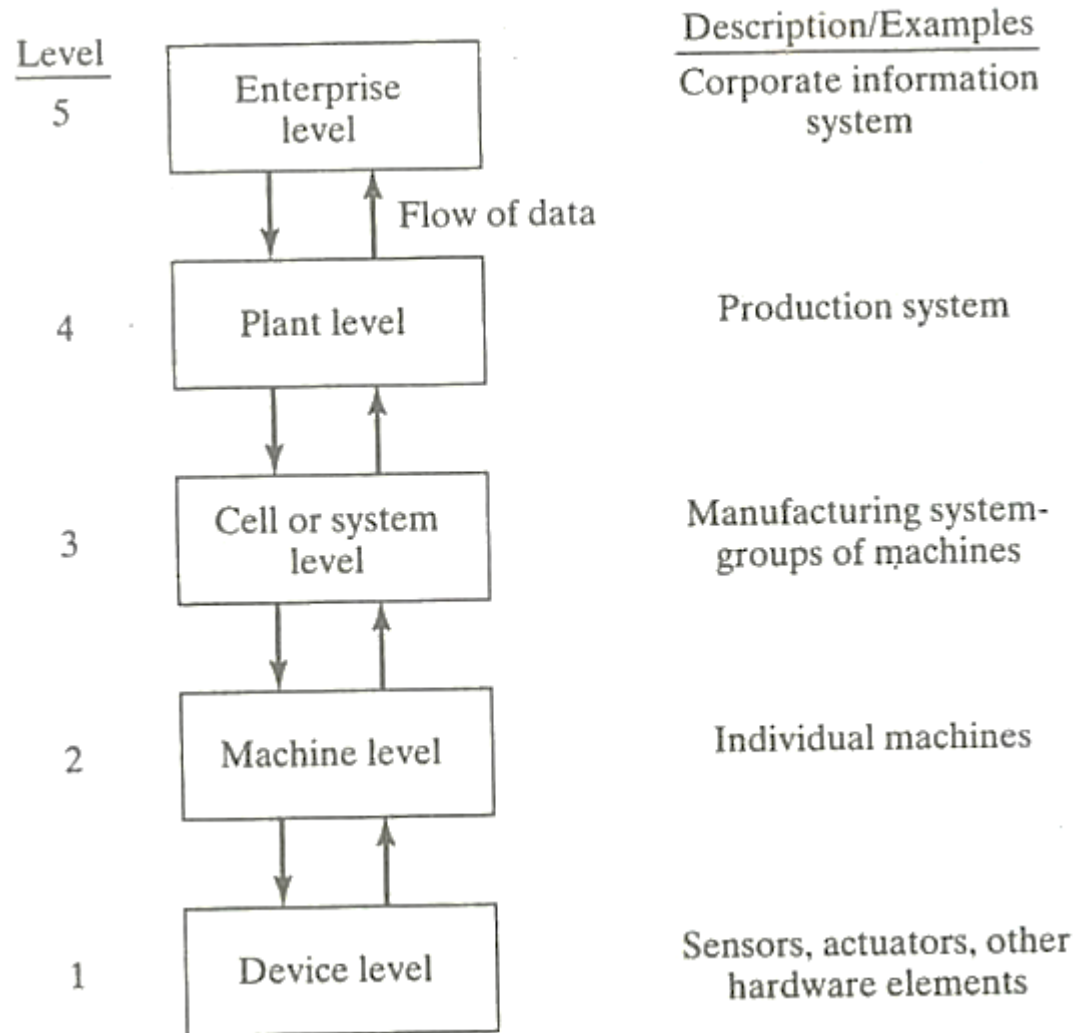
- **Level 3: Cell or system level.**
- It operates under instructions from the plant level.
- Manufacturing cell or system is a group of machines or workstations connected and supported by a material handling system, computer and other equipment appropriate to manufacturing process.
- Production lines are included in this level. Likely functions include part dispatching and machine loading, coordination among machines and material handling system, and collecting and evaluating inspection data

Hierarchical Structure of Industrial Automation



- **Level 4: Plant level.**
- Factory or production systems level.
- It receives instructions from (he corporate information system and translates them into operational plans for production.
- Likely functions include: order processing, process planning, inventory control, purchasing, material requirements planning, shop floor control, and quality control.

Hierarchical Structure of Industrial Automation



- **Level 5: Enterprise level.**
- Highest level consisting of the corporate information system.
- Concerned with all of the functions necessary to manage the company: marketing and sales, accounting, design, research, aggregate planning, and master production scheduling

Basic Components of Automation Systems



- Automation systems are the backbone of modern manufacturing, performing repetitive tasks with precision and efficiency.
- Input Devices (Sensors)
- Control Unit (Programmable Logic Controller - PLC)
- Output Devices (Actuators)

Basic Components of Automation Systems



- **Input Devices (Sensors):**
- Act as the eyes and ears of the system.
- Collect data about various physical parameters like temperature, pressure, flow rate, or position.
- Examples include temperature sensors in furnaces, pressure sensors in pipelines, and motion sensors for detecting object presence.

Input Devices (Sensors)



- Input devices, also known as sensors, act as the eyes and ears of the automation system.
- Primary function is to gather information about the physical environment surrounding the system.
- Sensors can measure various quantities, such as temperature, pressure, flow rate, position, presence/absence of objects, and many more.
- Convert these physical quantities into electrical signals that the control unit can understand and process.
- Wide variety of sensors exist for different applications, each with its own operating principles and characteristics.
- Common examples include temperature sensors, pressure sensors, flow sensors, limit switches, and photoelectric sensors.

Input Devices (Sensors) - Types



- **Temperature sensors:** Measure temperature (e.g., thermistors, thermocouples)
- **Pressure sensors:** Measure pressure or vacuum (e.g., piezometric sensors)
- **Flow sensors:** Measure the flow rate of fluids or gases (e.g., turbine flow meters, positive displacement flow meters)
- **Limit switches:** Detect the presence or absence of an object (e.g., mechanical switches)
- **Photoelectric sensors:** Detect the presence/absence of objects using light (e.g., photoelectric sensors)
- **Proximity sensors:** Detect presence or absence of an object without physical contact. (e.g., inductive proximity sensors for detecting metal objects and photoelectric sensors for detecting any object interrupting a light beam)

Signal Conditioning



- Raw electrical signals received from sensors may not always be directly usable by the PLC.
- Signals may be too weak, noisy, or require conversion into a format compatible with the PLC's input range.
- Signal conditioning is the process of preparing sensor signals for the PLC. This can involve:
- **Amplification:** If the sensor signal is weak, it can be amplified to a stronger level suitable for the PLC's input range.
- **Filtering:** Sensors may pick up electrical noise from the environment. Filtering removes this noise to ensure clean and accurate data for the PLC.
- **Scaling:** Sensor outputs may not directly correspond to the desired engineering units (e.g., temperature in degrees Celsius). Scaling converts the sensor output to a meaningful engineering unit for the PLC to interpret correctly.

Control Unit (PLC)



- Control unit is the brain of the automation system, responsible for making decisions and issuing commands.
- Programmable Logic Controllers (PLCs) are the most widely used type of control unit.
- Receive data from various sensors, interpret it based on pre-programmed logic instructions, and make control decisions.
- These decisions are then translated into electrical signals sent to actuators for physical action.
- PLCs typically offer a user interface for programming the control logic using specialized software and for monitoring the system's operation.

PLC Hardware Components



- **Central Processing Unit (CPU):** Acts as brain of the PLC, responsible for processing data received from sensors, executing the pre-programmed control logic, and making control decisions.
- **Input/Output (I/O) Modules:** Serve as bridge between PLC and physical world. Input modules receive electrical signals from sensors and convert them into a format CPU can understand while, output modules receive control signals from CPU and convert them into electrical signals that can activate actuators.
- **Power Supply:** Provides necessary electrical power to operate all PLC's internal circuits.
- **Programming Terminal:** Dedicated computer or software interface used for programming PLC's control logic, monitoring system operation, and troubleshooting any issues.

Output Devices (Actuators)



- **Output Devices (Actuators):**
- These are the physical movers and shakers of the system.
- Translate the control unit's instructions into physical actions.
- Examples include pneumatic cylinders that move objects, electric motors that drive machinery, and solenoid valves that control fluid flow.

Output Devices (Actuators)



- Output devices, also known as actuators, are the muscle of the automation system.
- Receive electrical signals from the control unit and convert them into physical actions that manipulate the environment.
- Actuators come in a wide variety to suit different applications. Here are some common types:
 - **Pneumatic actuators**
 - **Hydraulic actuators**
 - **Electric actuators**

Types of Actuators



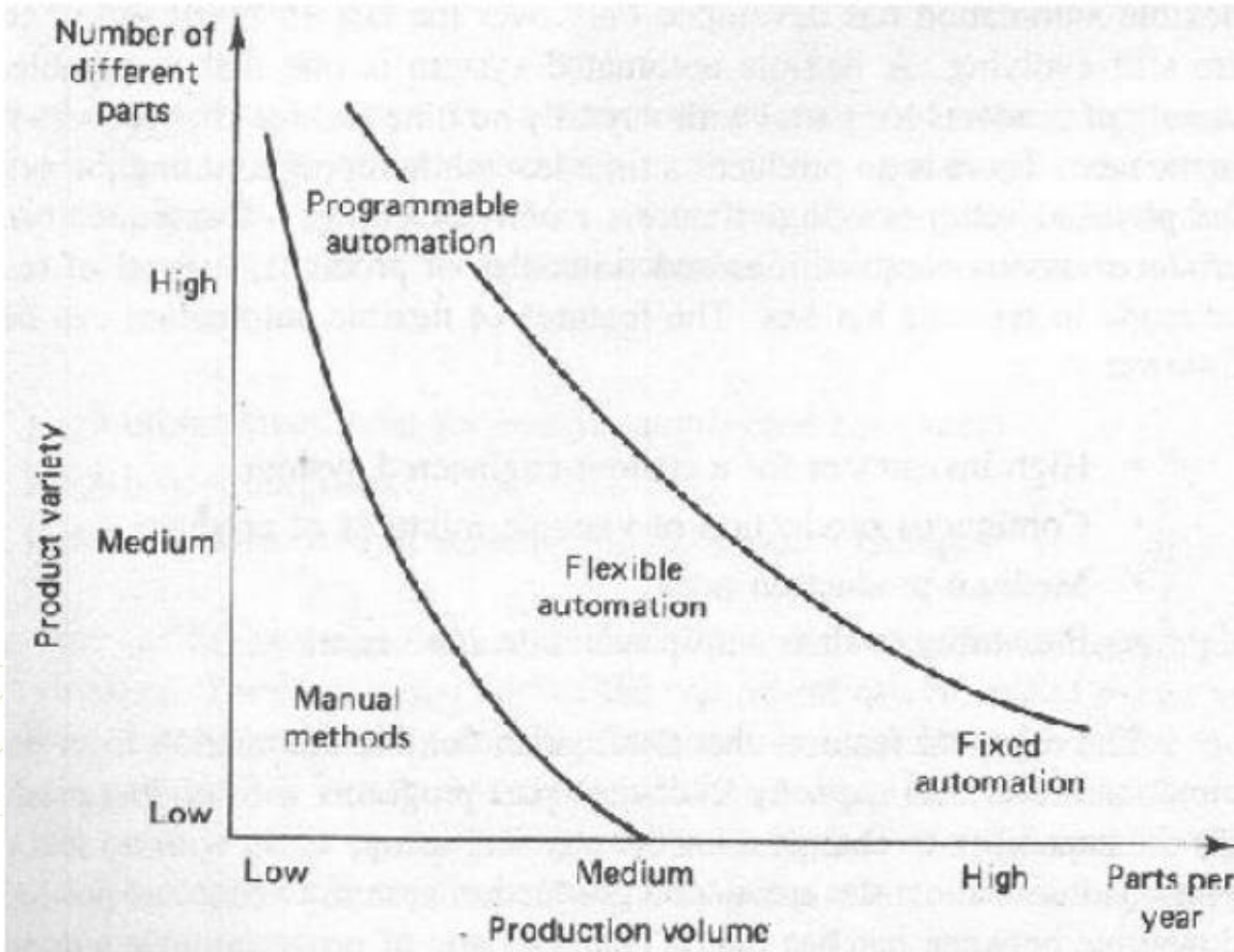
- **Pneumatic cylinders:** Use compressed air to generate linear motion for pushing, pulling, clamping, or lifting objects. Offer advantages like speed, simplicity, and cost-effectiveness for many applications.
- **Hydraulic actuators:** Use pressurized oil to generate linear or rotary motion for powering machinery or manipulating heavy objects. Provide high force and precise control but can be more complex and expensive than pneumatic systems.

Types of Actuators



- **Electric motors:** Convert electrical energy into rotary motion, driving various machines and equipment. Come in different types, such as AC motors for continuous rotation applications and DC motors for applications requiring variable speed control.
- **Solenoid valves:** Electronically controlled valves that regulate flow of fluids or gases in pipes. Offer fast response times and precise control over fluid flow, making them suitable for various applications.
- **Stepper motors** for precise positioning control and **linear motors** for high-speed linear motion.

Types of Automation Systems



- Automation plays a crucial role in various industries, transforming how tasks are performed and processes are controlled.
- Automation systems can be broadly categorized into three main types based on their flexibility and level of programmability:
 - fixed automation,
 - programmable automation, and
 - flexible automation.

Types of Automation Systems

- **Flexible Automation Systems**
 - **Definition:** Systems designed for rapid changeover between product types and production tasks.
 - **Examples:** CNC (Computer Numerical Control) machines, robotic assembly lines.
- **Fixed Automation Systems**
 - **Definition:** Systems designed for high-volume production of a specific product with minimal variation.
 - **Examples:** Automated conveyor systems in food processing plants, continuous production lines in automotive manufacturing.
- **Programmable Automation Systems**
 - **Definition:** Systems with a combination of fixed and flexible elements, allowing for customization through programming.
 - **Examples:** Automated packaging machines with adjustable settings, industrial robots programmed for various tasks.

Fixed automation

- Sequence of processing (or assembly) operations is fixed by equipment configuration.
- Operations in sequence are usually simple.
- Integration and coordination of many such operations into one piece of equipment that makes system complex.
- Economic justification for fixed automation is found in products with very high demand rates and volumes.
- High initial cost of equipment can be spread over a very large number of units, thus making unit cost attractive compared to alternative methods of production.
- **Typical features of fixed automation are:**
 - High initial investment for custom-engineered equipment
 - High production rates
 - Relatively inflexible in accommodating product changes

Fixed automation



- **Definition:** Fixed automation systems are designed for high-volume production of a single product or a limited range of similar products.
- **Characteristics:**
 - Limited flexibility
 - High initial investment
 - High production rates
 - Custom-designed equipment
 - Simple operation

Fixed automation Characteristics:

- **Limited flexibility:** The equipment is specifically designed for a particular product and cannot be easily adapted to produce different products.
- **High initial investment:** Due to the custom-designed nature of the equipment, fixed automation systems often involve a significant upfront cost.
- **High production rates:** Once set up and running, fixed automation systems can achieve high production rates due to their optimized and streamlined design.
- **Custom-designed equipment:** The machinery in fixed automation systems is typically custom-engineered to perform specific tasks for the product being produced.
- **Simple operation:** Once programmed, fixed automation systems require minimal operator intervention, making them relatively simple to operate.

Fixed Automation - Example: Automobile Assembly Line



- Entire production process, from stamping car parts to welding them together and painting the final car, is highly automated with fixed equipment.
- Each station on the assembly line performs a specific task, and the entire line is designed for high-volume production of a particular car model.
- Reconfiguring such a line for a different car model would require significant modifications and potentially new equipment.



- Provide a balance between flexibility and production efficiency.
- Offer more flexibility than fixed automation by allowing for production of a variety of products within a certain product family.
- Can be programmed to produce a variety of products with different configurations within a certain product family.
- **Characteristics:**
 - Moderate flexibility
 - Moderate initial investment
 - Programmable control units
 - Adaptable to product variations

Programmable Automation characteristics:



- **Moderate flexibility:** Programmable systems can be adapted to produce different product variations by changing the program in the control unit.
- **Moderate initial investment:** Compared to fixed automation, programmable systems typically require a lower initial investment due to their adaptability.
- **Programmable control units:** These systems utilize programmable logic controllers (PLCs) that allow for storing and executing control programs for different product variations.
- **Adaptable to product variations:** By modifying the program in the PLC, programmable systems can be adapted to produce different versions of a product with varying features or configurations.



- Example of programmable automation is Computer Numerical Control (CNC) machining.
- CNC machines are equipped with programmable controllers that allow them to be used for various machining operations like drilling, milling, and turning.
- By changing the program in the controller, a CNC machine can be used to produce different parts with varying dimensions and features.



- Offer highest level of flexibility amongst the three types.
- Can be easily reconfigured to produce a wide range of entirely different products.
- **Characteristics:**
 - High flexibility
 - Higher initial investment (compared to programmable automation)
 - Modular design
 - Reconfigurable for different products
 - Suitable for low- to medium-volume production

- **High flexibility:** Excel in their ability to be adapted to produce various products with minimal hardware modifications.
- **Higher initial investment** (compared to programmable automation): While offering more flexibility, flexible automation systems typically require a higher initial investment due to their modular design and reconfiguration capabilities.
- **Modular design:** Built using modular components that can be easily rearranged and reprogrammed to accommodate different product requirements.
- **Reconfigurable for different products:** Allows for quick and efficient reconfiguration of the system to produce new products.
- **Suitable for low- to medium-volume production:** Most suited for production scenarios with lower or medium product volumes due to the time and resources involved in reconfiguration.

Flexible Automation - Example: Robotic Assembly Line



- Robotic assembly line employing robots with reprogrammable arms and grippers is a prime example of flexible automation.
- Robots can be programmed to perform various assembly tasks, such as picking and placing components, welding, and fastening.
- Entire line can be reconfigured to assemble different products by modifying the robot programs and potentially rearranging the layout.

- **Production volume:** For high-volume production of a single product, fixed automation offers the most efficient and cost-effective solution.
- **Product variety:** If you need to produce a range of similar products with some variations, then programmable automation provides the flexibility to adapt the system through program changes.
- **Product lifecycle:** For products with frequent design changes or short lifecycles, flexible automation allows for easier reconfiguration to accommodate these changes.
- **Budgetary constraints:** Fixed automation often requires a higher initial investment due to the custom-designed equipment. Programmable and flexible systems offer varying levels of upfront costs depending on their complexity.

- **Increased productivity:** Automation systems can operate continuously with minimal downtime, leading to significant increases in production output.
- **Improved quality:** Automated processes ensure consistent quality by eliminating human error and maintaining precise control over parameters.
- **Reduced costs:** While the initial investment may be high, automation can lead to cost savings in the long run through reduced labor costs, lower scrap rates, and increased efficiency.
- **Enhanced safety:** Automation can remove human workers from hazardous environments, reducing the risk of workplace accidents and injuries.

Introduction to the Future Automated Factory



- Interaction between robots and human is essential factor to application of robots in future.
- Potential to change both society and the economy in a positive direction since it could build a bridge between machines and humans.
- Critical aspects of robots is their ability to interact with environment by making their own decisions.
- These critical judgments can be made from the robots observing and sensing the environment, which allows them to build social interactions with human workers.

Introduction to the Future Automated Factory



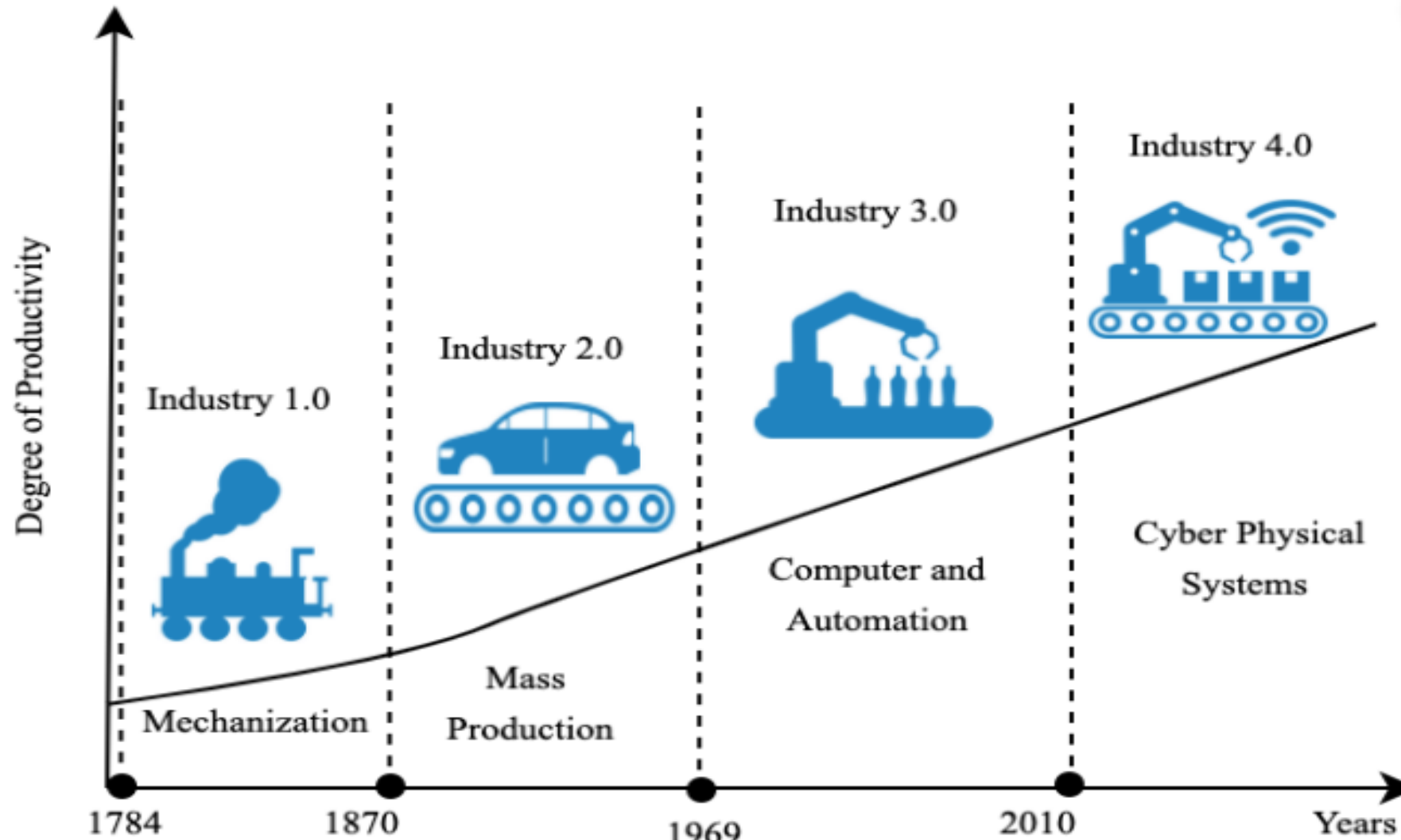
- Concern is how to implement computational algorithms in a manner that it will be safe for workers to work around robots.
- Due to risks of exposing human workers to malfunctioned machines, nowadays, robots are mostly separated from humans to ensure safety, and this leads to a decrease in both their flexibility and productivity.
- Attention has been concentrated at robot-human interaction for production and manufacturing purposes.

What is a Future Automated Factory?



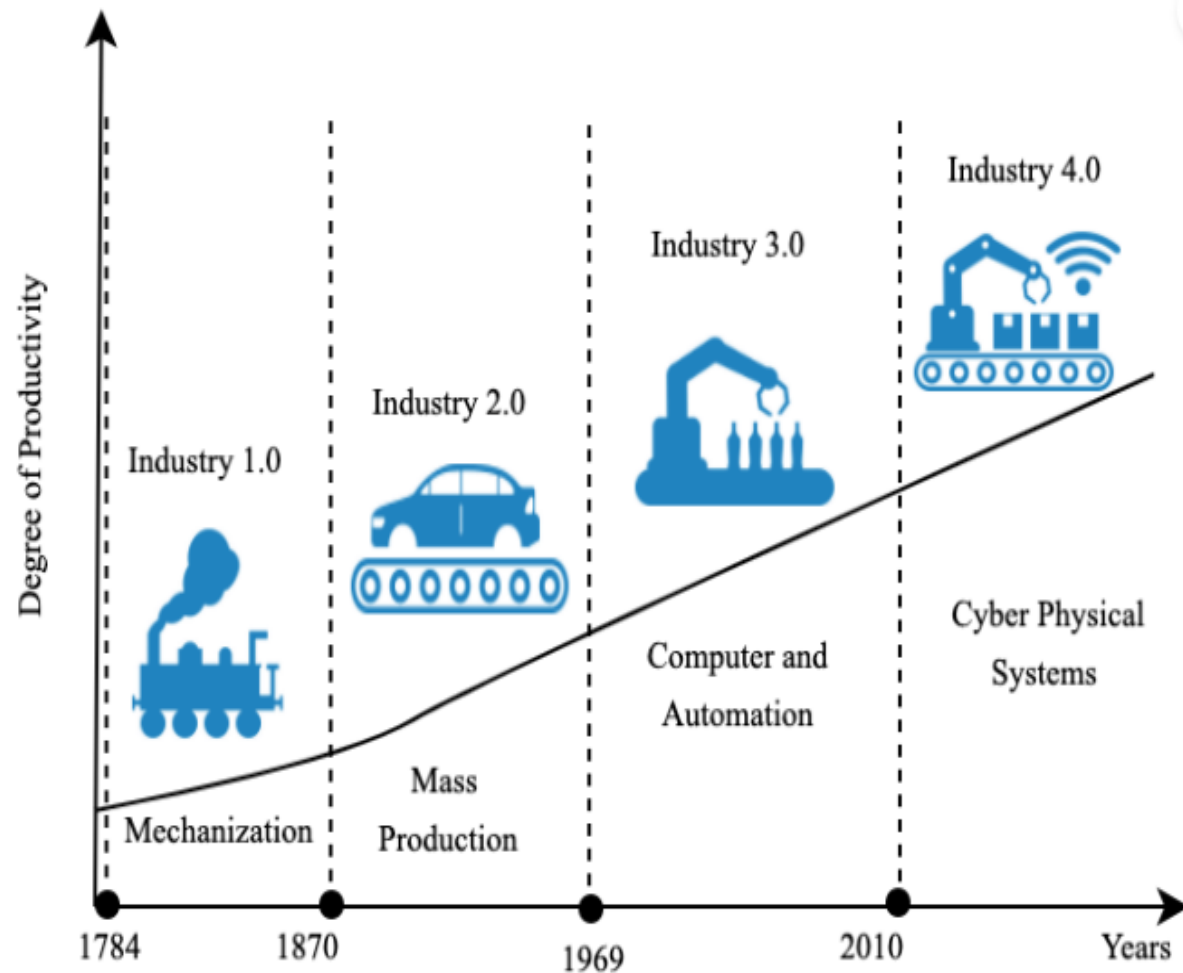
- A future automated factory is a highly advanced manufacturing facility that leverages automation technologies to a significant degree.
- These factories utilize sophisticated robots, intelligent machines, and interconnected systems to perform tasks traditionally done by human workers.
- Automation offers several advantages, including increased efficiency due to continuous operation, improved product quality through consistent control, and greater flexibility to adapt to changing production demands.

Evolution of Manufacturing Technologies



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Evolution of Manufacturing Technologies



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- Concept of automation in manufacturing has a rich history.
- Seeds were sown during Industrial Revolution, which saw the rise of mechanized production using water and steam power.
- This era laid foundation for increased production efficiency.
- Concept was further revolutionized by Henry Ford's assembly line concept, which introduced a standardized and streamlined approach to production.
- Late 20th century witnessed a significant breakthrough with development of industrial robots, paving way for a more automated future in manufacturing.

Key Drivers of Automation in Manufacturing



- Trend towards automation in manufacturing is fueled by several key drivers.
- **Increasing labor costs:** Rising labor costs in developed countries make automation a more cost-effective solution.
- **Global competition:** Manufacturers need to be efficient and adaptable to compete in the global market.
- **Technological advancements:** Advancements in robotics, artificial intelligence, and machine learning are making automation more sophisticated and affordable.
- **Demand for customization:** Consumers are demanding more customization, which can be challenging with traditional manufacturing methods.

Key Features of Future Automated Factories

Future automated factories boast several key features that differentiate them from traditional manufacturing facilities.

- **Highly automated production:** Robots and intelligent machines perform most production tasks, with minimal human intervention.
- **Advanced robotics:** Sophisticated robots with increased dexterity, vision capabilities, and AI-powered decision-making handle complex tasks.
- **Interconnected systems:** Machines and devices communicate with each other through networks, enabling real-time data exchange and coordinated operations.
- **Data-driven decision making:** Production processes are optimized based on data collected from sensors and intelligent systems.

Components of a Future Automated Factory



- **Industrial robots:** Workhorses, perform various tasks like welding, assembly, material handling, and inspection.
- **Intelligent machines:** Equipped with sensors and AI for tasks requiring decision-making and adaptation; no longer dumb machines
- **Conveyor belts and automated guided vehicles (AGVs):** Efficiently transport materials and products throughout the factory.
- **Sensors and monitoring systems:** Continuously collect data on machine performance, product quality, and environmental conditions.
- **Central computer system:** Acts as the brain of the factory, controlling operations, analyzing data, and optimizing processes.

Challenges of Future Automated Factories

- **High initial investment:** Setting up a future automated factory requires a significant upfront investment in advanced robotics, intelligent machines, and interconnected systems.
- **Job displacement:** Automation may lead to job losses in the manufacturing sector as robots and machines replace human workers in many tasks.
- **Technical complexity:** Maintaining and troubleshooting complex automated systems requires skilled personnel with expertise in robotics, AI, and data analysis.
- **Security risks:** Interconnected systems and reliance on central computers raise concerns about cybersecurity vulnerabilities and potential disruptions.

Technological Advancements in Automation



- Exciting world of technological advancements shaping the future of manufacturing automation.
- Cutting-edge technologies like robotics, artificial intelligence, and the Internet of Things (IoT) that are transforming how products are designed, manufactured, and delivered.

Cutting-Edge Technologies Shaping Manufacturing



- Future of manufacturing is being shaped by a convergence of powerful technologies.
- Advancements in robotics, with industrial robots and collaborative robots (cobots) are taking on an increasingly significant role in production processes.
- Artificial intelligence and machine learning are revolutionizing how machines learn, adapt, and optimize operations.
- Interconnectedness of machines and devices through the Internet of Things (IoT) and Industrial IoT (IIoT) allows for real-time data collection and analysis, enabling smarter decision-making.
- 3D printing and additive manufacturing are transforming the way we create objects, offering greater design flexibility and on-demand production capabilities.
- These technologies, working together, are creating a future of intelligent, flexible, and efficient manufacturing.

Robotics and Cobots



- Robots have long played a vital role in automation, and advancements in robotics continue to push the boundaries.
- Industrial robots are the workhorses of manufacturing, performing tasks with unmatched speed, precision, and tireless consistency.
- These robots are increasingly sophisticated, equipped with advanced sensors and vision systems for greater flexibility and adaptability.
- Emergence of collaborative robots, or cobots.
- These robots are specifically designed to work safely alongside human workers, assisting with repetitive tasks, material handling, and quality inspection.
- Cobots are typically smaller and more lightweight than industrial robots, and their user-friendly design allows for easier integration into existing workspaces.



- AIML are transforming manufacturing by enabling machines to learn, analyze data, and make intelligent decisions.
- AI algorithms can analyze vast amounts of data from sensors and machines to identify patterns, predict trends, and recommend adjustments to optimize production processes for efficiency and quality. This data can include information on machine performance, product quality, and environmental conditions.
- Machine learning algorithms allow machines to learn and improve their performance over time without explicit programming. For example, a machine learning system can analyze data on welding processes and identify ways to optimize parameters for stronger and more consistent welds.

IoT and Industrial IoT (IIoT)



- IoT and Industrial IoT (IIoT) are revolutionizing the way machines communicate and share data.
- Internet of Things (IoT) connects devices and machines to the internet, enabling real-time data collection and communication.
- Industrial IoT (IIoT) focuses on connecting industrial machines and devices for data exchange and coordinated operations within a manufacturing environment.
- IoT essentially connects everyday devices to the internet, allowing them to collect and transmit data. In an industrial setting, this translates to Industrial IoT

3D Printing and Additive Manufacturing



- 3D printing, also known as additive manufacturing, creates three-dimensional objects by depositing material layer by layer.
- Offers greater design flexibility and the ability to create complex geometries that are difficult or impossible with traditional manufacturing methods.
- Enables on-demand production, reducing the need for large inventories and allowing for faster customization.
- Technology is particularly well-suited for prototyping, creating complex parts for machines, and even printing medical implants customized for individual patients.

Case Studies: Advanced Automated Factories

- Not many large-scale, fully automated factories in India compared to some developed nations, several Indian companies are actively adopting automation technologies to improve their manufacturing processes.
- Examples include:
- **Case 1: Tata Motors' Pune Plant**
- **Company:** Tata Motors (Automobile Manufacturing)
- **Technology Focus:** Robotics and Automation
- **Implementation:** Tata Motors' Pune plant leverages a high degree of automation in its car assembly lines. Robots perform tasks like welding, painting, and engine assembly with precision and speed. This automation has resulted in increased production efficiency and improved product quality.

Case 2: Bharat Forge's Chakan Plant

- **Company:** Bharat Forge (Automobile Components)
- **Technology Focus:** Robotics, Cobots, and IIoT
- **Implementation:** Bharat Forge's Chakan plant utilizes a combination of industrial robots, collaborative robots (cobots), and Industrial IoT (IIoT) for forging, machining, and quality control processes. Robots handle heavy-duty tasks, while cobots work alongside human workers for tasks requiring dexterity and human judgment. IIoT enables real-time monitoring of machine performance and predictive maintenance, minimizing downtime

Case 3: Aditya Birla Group's Grasim Industries

- **Company:** Grasim Industries (Textile Manufacturing)
- **Technology Focus:** Automation and Data Analytics
- **Implementation:** Grasim Industries has implemented automated spinning and weaving processes in its textile manufacturing facilities. Additionally, they utilize data analytics to optimize production parameters, reduce waste, and ensure consistent product quality.

Case 4: Maruti Suzuki India

- **Company:** Maruti Suzuki India (Automobile Manufacturing)
- **Technology Focus:** Automation and Skill Development
- **Implementation:** Maruti Suzuki, a leader in the Indian automobile industry, has extensively adopted automation in its production lines. However, they acknowledge the importance of human expertise. The company invests heavily in skill development programs to ensure their workforce can effectively manage and maintain the automated systems.

Human Workers in Future Automated Factory



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- Rise of automation is transforming future of manufacturing.
- Robots are increasingly taking over repetitive tasks, human workers remain crucial for the success of future automated factories.
- **What's the role of human workers in this new industrial landscape.**

Role of Human Workers in Future Automated Factories

Future of work in manufacturing doesn't involve robots replacing humans entirely, instead human-machine collaboration.

Human workers will assume new and more sophisticated roles, focusing on tasks that leverage their unique capabilities.

- **Supervision and oversight:** Human workers will play a critical role in supervising automated systems, monitoring performance, and identifying potential issues.
- **Decision-making and problem-solving:** Complex decision-making, troubleshooting technical problems, and adapting to unforeseen circumstances will remain human strengths.
- **Quality control and inspection:** Ensuring product quality through inspections and adjustments requires human judgment and keen observation skills.
- **Data analysis and process optimization:** Analyzing data collected from sensors and intelligent systems to identify areas for improvement requires human expertise.
- **Creativity and innovation:** Human creativity will be vital for developing new products, processes, and ways to leverage automation effectively.

Transition from Manual Labor to Skilled Labor

- Transition necessitates change in skillset required for workers in manufacturing sector.
- Repetitive manual labor tasks will be increasingly automated, so future workforce will need to be more skilled and adaptable.
- Workers will need training in areas like:
 - Robotics and automation technologies
 - Data analysis and interpretation
 - Programming and maintenance of automated systems
 - Advanced manufacturing processes
 - Human-machine interface design
 - Problem-solving and critical thinking

Importance of Human-Machine Collaboration

- Human-machine collaboration is vital for optimizing production processes and achieving the full potential of future automated factories.
- Humans excel at:
 - Decision-making and critical thinking
 - Creativity and innovation
 - Adaptability and problem-solving
 - Communication and collaboration
 - Social and emotional intelligence

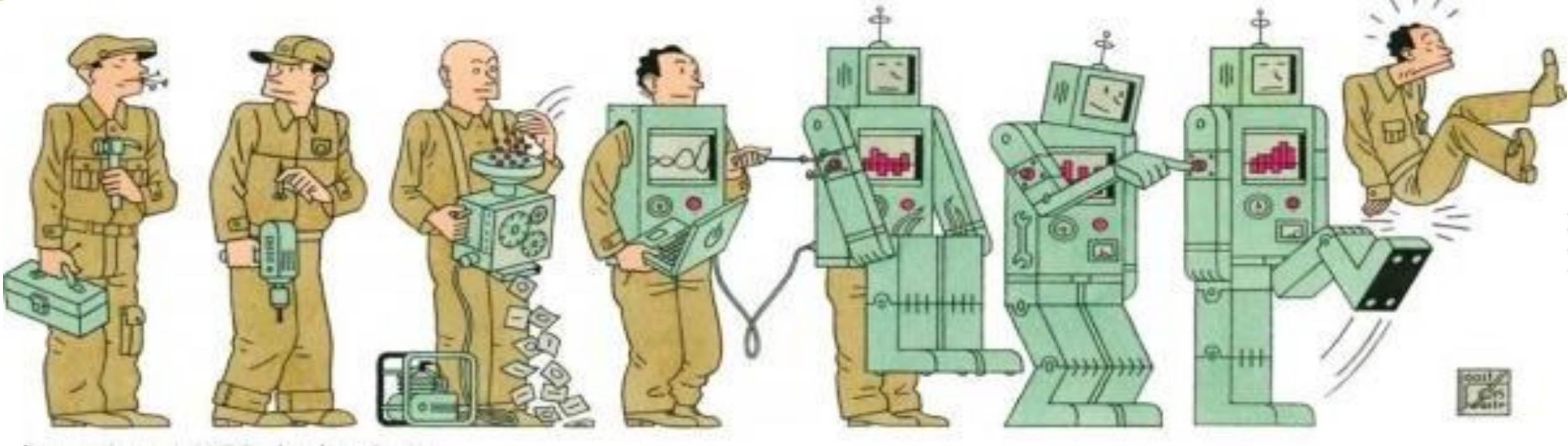
Human-Machine Interface Design

- User-friendly human-machine interface (HMI) design is crucial for effective human-machine collaboration.
- HMI acts as the bridge between humans and machines, allowing them to interact and exchange information effectively.
- Key considerations for HMI design:
 - Intuitive and easy-to-use interface
 - Clear and concise information display
 - Ability to provide feedback and control
 - Compatibility with human cognitive abilities
 - Training and support for workers

Training and Upskilling Programs for Workers

- Transition towards automation necessitates ongoing training and upskilling programs for existing workforce.
- Programs should equip workers with knowledge and skills required to thrive in a future dominated by automated systems.
- Training programs should focus on:
 - Robotics and automation technologies
 - Data analysis and interpretation
 - Programming and maintenance of automated systems
 - Problem-solving and critical thinking skills
 - Human-machine collaboration and communication

Job Displacement vs. Job Enhancement



- Potential job displacement can occur in repetitive, manual labor tasks.
- While automation can displace some jobs, it's also creating new job opportunities in several areas.
- Development, deployment, and maintenance of these automated systems require a skilled workforce.

Job Displacement vs. Job Enhancement

- Robotics engineers and technicians are needed to design, build, and maintain these sophisticated robots.
- Data analysts and system specialists are crucial for extracting insights from vast amount of data generated by automated systems.
- As human-machine collaboration becomes increasingly important, HMI designers and user experience specialists will be in high demand to create user-friendly interfaces that facilitate effective interaction.
- Automation specialists and process optimization experts will be needed to analyze existing processes, identify areas for improvement, and leverage automation to achieve greater efficiency and productivity.

The Social Impact of Automated Factories



- Automation is transforming the manufacturing sector, and the implications extend far beyond factory walls.
- What are the economic effects on employment and wages, the changing skill requirements for the workforce, considerations for workplace safety and ergonomics, and broader societal challenges and opportunities.
- What are the ethical considerations surrounding privacy, job security, and income inequality, and explore strategies for mitigating negative impacts and promoting inclusive growth in a future shaped by automation.

Economic Implications of Automation



- **Potential for job displacement:** Automation can certainly replace human workers in tasks that are repetitive, predictable, and require minimal cognitive skills.
- **Impact on wages:** Automation could put downward pressure on wages for some occupations while increasing wages for skilled workers in areas like robotics and data analysis.
- **Increased productivity and economic growth:** Automation can lead to increased production efficiency and economic growth, potentially creating new job opportunities in other sectors.

- **Shift towards higher-skilled jobs:** Automation will likely necessitate a workforce with a more advanced skillset.
- **Demand for skills in areas like:**
 - Robotics and automation technologies
 - Data analysis and interpretation
 - Programming and maintenance of automated systems
 - Problem-solving and critical thinking
 - Human-machine collaboration

- Safety is paramount in any work environment, and automated factories present unique considerations.
- **Safety considerations with robots and automated systems:** Proper safety protocols and training are essential to minimize risks.
- **Ergonomic design for human-machine interaction:** Workstations and interfaces should be designed to prevent repetitive strain injuries and ensure worker comfort.
- **Importance of a safety culture:** Fostering a culture of safety awareness is crucial for preventing accidents and promoting well-being in the workplace.

- **Challenges:**

- Potential for increased unemployment and income inequality
- Widening skills gap between those who can adapt and those who struggle
- Devaluation of certain types of work and potential social unrest

- **Opportunities:**

- Increased productivity and economic growth
- Creation of new jobs in different sectors
- Potential for improved working conditions with automation taking over hazardous tasks
- Opportunity to focus on upskilling and reskilling the workforce for a more diverse range of jobs

Ethical Considerations: Privacy, Job Security, and Income Inequality

- **Privacy concerns:** Automated systems collect vast amounts of data, including data on worker productivity, movement patterns, and even machine interactions. There's a need for robust data privacy policies and regulations to ensure this data is collected ethically, used responsibly, and protected from unauthorized access.
- **Job security anxieties:** Fear of job displacement due to automation can lead to anxiety and stress among workers. Companies have a responsibility to support their workforce through this transition by providing training and reskilling opportunities.
- **Income inequality:** Automation could exacerbate income inequality if the benefits primarily accrue to those who own and develop the technology.

Strategies for Mitigating Negative Social Impacts and Promoting Inclusive Growth

- **Investment in education and training:** Upskilling and reskilling programs are crucial to prepare the workforce for the future.
- **Social safety nets:** Policies to support displaced workers and ensure they have access to retraining and income security.
- **Universal basic income (UBI):** A potential solution to address income inequality in a world with widespread automation.
- **Lifelong learning initiatives:** Encouraging a culture of continuous learning to adapt to the changing demands of the workforce.
- **Ethical regulations on data collection and use:** Ensuring responsible data practices to protect worker privacy.

Fourth Unit Complete

- **UNIT IV**
- **Introduction to Industrial Automation:** Need and benefits of Industrial Automation, Automation Hierarchy, Basic components of Automation systems, description of each component, types of automation systems flexible, fixed and programmable systems
- **The Future Automated Factory:** Future Automated Factory, Human Workers in the Future Automated Factory, The social impact